

KOBE, Japan

WMO: 477700

Lat: 34.6969N

Lon: 135.2128E

Elev: 31

StdP: 100.96

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.2	(c) 1.2	(d) -8.5	(e) 1.8	(f) 3.3	(g) -7.0	(h) 2.1	(i) 4.0	(j) 11.0	(k) 5.8	(l) 9.7	(m) 6.2	(n) 4.3	(o) 280	(p) 0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.7	(c) 33.4	(d) 25.3	(e) 32.3	(f) 25.0	(g) 31.3	(h) 24.8	(i) 26.6	(j) 30.6	(k) 26.2	(l) 30.1	(m) 25.8	(n) 29.6	(o) 4.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	28.8	25.0	20.2	28.7	24.6	19.7	28.5	83.2	30.7	81.5	30.2	79.8	29.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.4	(c) 7.3	(e) -1.8	(f) 35.4	(g) 1.6	(h) 1.2	(i) -3.0	(j) 36.2	(k) -3.9	(l) 36.9	(m) -4.8	(n) 37.6	(o) -5.9	(p) 38.4		
(a) 9.9	(b) 8.4	(c) 7.3	(e) -3.8	(f) 27.2	(g) 1.5	(h) 0.7	(i) -4.8	(j) 27.7	(k) -5.7	(l) 28.1	(m) -6.5	(n) 28.5	(o) -7.6	(p) 29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.0	6.5	9.6	15.0	20.0	23.5	27.5	29.0	25.5	20.0	14.0	8.7
	DBStd	8.34	2.28	2.78	3.09	3.18	2.29	1.98	2.14	1.70	2.64	2.74	3.06	2.81
	HDD10.0	340	125	103	46	3	0	0	0	0	0	0	3	59
	HDD18.3	1548	382	331	269	106	11	0	0	0	0	14	133	300
	CDD10.0	2957	2	6	34	153	309	404	543	590	466	309	123	18
	CDD18.3	1121	0	0	0	6	61	154	285	332	216	65	3	0
	CDH23.3	9723	0	0	0	7	137	724	2923	3946	1801	184	1	0
	CDH26.7	3227	0	0	0	0	10	97	981	1644	482	12	0	0
Wind	WSAvg	3.4	3.6	3.4	3.5	3.4	3.4	3.3	3.2	3.5	3.6	3.5	3.3	3.6
Precipitation	PrecAvg	1315	42	63	99	102	142	179	163	104	165	120	82	54
	PrecMax	1720	100	148	176	185	371	382	430	385	421	317	192	104
	PrecMin	672	5	16	41	27	33	57	26	15	41	32	17	4
	PrecStd	283	24	35	32	41	76	76	108	88	88	68	44	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	16.8	19.2	24.2	27.9	30.4	34.7	35.2	32.8	28.1	22.4	17.6
		MCWB	11.2	14.0	13.5	15.8	18.4	22.7	25.8	25.4	24.6	22.1	17.0	14.3
	2%	DB	12.1	14.4	17.6	22.2	26.0	28.8	32.9	33.9	31.4	26.5	21.0	15.5
		MCWB	8.4	10.7	12.8	15.4	17.9	22.1	25.2	25.4	24.3	20.9	16.5	11.7
	5%	DB	10.8	12.6	16.3	21.0	24.8	27.7	31.7	32.8	30.3	25.3	19.9	14.1
		MCWB	7.3	9.1	12.1	15.2	17.7	21.7	25.0	25.2	23.9	19.9	15.7	10.3
	10%	DB	9.7	11.1	14.9	19.7	23.7	26.7	30.6	31.9	29.3	24.2	18.7	13.0
		MCWB	6.3	7.7	10.9	14.7	17.5	21.2	24.8	25.0	23.5	18.8	14.6	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	14.5	15.9	19.5	21.7	25.1	26.9	27.2	26.6	24.0	19.3	15.4
		MCDB	13.9	16.3	17.3	21.2	23.4	28.1	31.8	31.4	30.0	26.2	20.6	17.1
	2%	WB	9.5	11.7	14.1	17.9	20.8	24.3	26.2	26.7	25.7	22.2	17.6	12.6
		MCDB	11.2	13.7	16.5	20.0	23.0	26.9	30.5	30.7	29.3	25.4	20.0	14.7
	5%	WB	7.9	9.8	12.7	16.7	19.8	23.7	25.8	26.4	25.2	20.9	16.4	11.0
		MCDB	10.3	11.8	15.5	19.6	22.5	26.1	29.8	30.3	28.8	24.2	19.0	13.6
	10%	WB	6.5	8.0	11.4	15.6	18.9	22.8	25.5	26.1	24.6	19.7	15.2	9.6
		MCDB	9.4	10.8	14.5	18.9	22.2	25.3	29.3	29.9	28.3	23.3	18.3	12.5
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	6.8	7.0	6.7	5.7	5.3	5.7	5.7	6.0	6.0	5.7
		MCDBR	6.6	7.4	7.8	7.9	8.0	6.8	6.5	6.7	6.3	6.5	6.4	6.5
		MCWBR	5.1	5.9	6.0	4.6	4.2	3.2	2.4	2.6	3.1	4.1	4.9	5.0
	5% WB	MCDBR	6.2	6.6	7.0	6.4	5.8	4.8	5.7	5.6	5.3	5.6	5.8	6.1
		MCWBR	5.7	6.3	6.7	5.2	4.5	2.6	2.3	2.4	3.0	4.6	5.4	5.7
Clear-Sky Solar Irradiance	taub	0.415	0.459	0.525	0.557	0.562	0.583	0.585	0.552	0.503	0.461	0.432	0.409	
	taud	2.101	2.000	1.857	1.832	1.884	1.904	1.949	2.026	2.127	2.160	2.141	2.130	
	Ebn at Noon	756	765	748	748	750	733	728	745	761	758	734	734	
	Edh at Noon	125	154	193	208	200	195	186	169	146	130	120	115	
All-Sky Solar Radiation	RadAvg	2.24	2.87	3.88	4.76	5.21	4.72	4.99	5.13	4.01	3.25	2.45	2.03	
	RadStd	0.16	0.23	0.30	0.42	0.52	0.44	0.66	0.48	0.35	0.32	0.20	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page